

How do we find approximate solution to overdetermined system?

no. of eq, $m >$ no. of unknown, n

* Discrete Least Square

$$Ax = b$$

[m x n]

multiplying with A^T

$$A = [m \times n]$$

$$A^T = [n \times m]$$

$$A^T \cdot A$$

$$n \times n$$

$$A^T A x = A^T b$$

Example: Fit a least square straight line (degree=1)

to the given data $f(-3) = f(0) = 0$, $f(6) = 2$

$\Rightarrow$ node = 3, degree = 2

$$P_2(x) = a_0 + a_1 x + a_2 x^2$$

We need to find $P_1(x) = a_0 + a_1 x$

$$P_1(-3) = a_0 + a_1(-3) = 0$$

$$P_1(0) = a_0 + a_1(0) = 0$$

$$P_1(6) = a_0 + a_1(6) = 2$$

Coeff matrix, $A = \begin{pmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{pmatrix}$, $b = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$

Now,

$$A^T A x = A^T b$$

$$\Rightarrow \begin{pmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -3 & 0 & 6 \end{pmatrix} \begin{pmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ -3 & 0 & 6 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 3 & 3 \\ 3 & 45 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 2 \\ 12 \end{pmatrix}$$

Using Gaussian to solve,

$$\text{Aug}(A) := A | b = \begin{pmatrix} 3 & 3 & | & 2 \\ 3 & 45 & | & 12 \end{pmatrix}$$

1st row operation,

$$m_{21} = \frac{a_{21}}{a_{11}} = 1$$

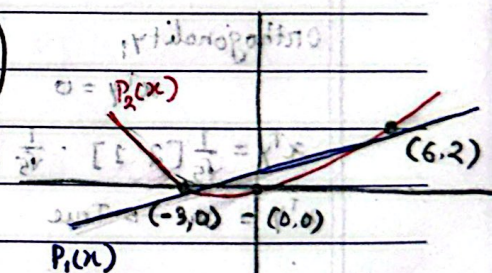
$$\text{New Aug}(A) = \begin{pmatrix} 3 & 3 & | & 2 \\ 0 & 42 & | & 10 \end{pmatrix} \equiv (Ux = b')$$

$$r_1 \rightarrow r_1 = (3 \ 3 \ 2)$$

$$r_2 \rightarrow m_{21} r_1 = r_2 - m_{21} r_1 = (0 \ 42 \ 10)$$

$$a_0 = \frac{3}{7}, \quad a_1 = \frac{5}{21}$$

Least square straight line: $P_1(x) = \frac{3}{7} + \frac{5}{21}x$
approximation



Orthonormality:

A set of vectors that are both orthogonal & Normal

Row Matrix

Orthogonal

Orthonormal

Every pair of vectors are perpendicular.

Each vector has unit length (magnitude = 1)



(Dot Product = 0)

$$\begin{matrix} x^T y = 0 \\ y^T z = 0 \\ \vdots \end{matrix}$$

$$\begin{matrix} x^T x = 1 \\ y^T y = 1 \\ z^T z = 1 \\ \vdots \end{matrix}$$

Example: Check if the following vector set is orthonormal.

$$S = \left\{ \frac{1}{\sqrt{5}} [2, 1]^T, \frac{1}{\sqrt{5}} [1, -2]^T \right\}$$

Orthogonality,

$$x^T y = 0$$

$$x^T y = \frac{1}{\sqrt{5}} [2 \ 1] \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$x^T y = 0 \rightarrow \text{True}$$

Normality,

$$x^T x = 1$$

$$x^T x = \frac{1}{\sqrt{5}} [2 \ 1] \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$x^T x = 1 \rightarrow \text{True}$$

Since both orthogonality and normality has been proved,

So, the matrix has orthonormal property.

$$x \frac{1}{\sqrt{5}} + \frac{1}{\sqrt{5}} = (x)^T$$

6.2 : QR Decomposition

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$A = QR$$

Orthonormal
set of vectors

Upper triangular Matrix, $R = Q^T A$

$$Q = [q_1 \ q_2 \ \dots \ q_n]$$

$$R = \begin{bmatrix} u_1^T q_1 & u_1^T q_2 & \dots & u_1^T q_n \\ u_2^T q_1 & u_2^T q_2 & \dots & u_2^T q_n \\ 0 & u_2^T q_2 & \dots & u_n^T q_2 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & u_n^T q_n \end{bmatrix}$$

To find Q and R we use Gram-Schmidt Process.

To find q_k , we need two formulas:

$$p_k = u_k - \sum_{i=1}^{k-1} (u_k^T q_i) q_i$$

$$q_k = \frac{p_k}{|p_k|}$$

QR Decomposition Formula derivation:

$$Ax = b$$

from $\Rightarrow A^T A x = A^T b$ [From Discrete least square method]

$$\Rightarrow (QR)^T QR x = (QR)^T b \quad [\text{Since } A = QR]$$

$$\Rightarrow Q^T R^T Q R x = Q^T R^T b$$

$$\Rightarrow Q^T Q R^T R x = Q^T R^T b$$

$$\Rightarrow R^T R x = Q^T R^T b \quad [Q \text{ is orthonormal}]$$

$$\text{So, } Q^T Q = I$$

$$\Rightarrow \boxed{Rx = Q^T b}$$

Solve Steps:

1. Using Gram Schmidt Process to find $q_1, q_2, \dots, q_n$ [n = no. of unknowns]
2. $Q = [q_1, q_2, \dots, q_n]$
3. Find R
4. Solve $Rx = Q^T b$

Ex Example: Fit a straight line using QR decomposition for given data:

$$f(-3) = f(0) = 0, f(6) = 2$$

$$P_1(x) = a_0 + a_1 x$$

$$P_1(-3) = a_0 + a_1(-3) = 0$$

$$P_1(0) = a_0 + a_1(0) = 0$$

$$P_1(6) = a_0 + a_1(6) = 2$$

$$A = \begin{pmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{pmatrix} \quad \text{let, } u_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad u_2 = \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix}$$

Step 1: $P_k = u_k - \sum_{i=1}^{k-1} (u_k^T q_i) q_i$

$$q_k = \frac{P_k}{|P_k|}$$

$k=1,$

$$P_1 = u_1 - 0 \quad q_1 = \frac{P_1}{|P_1|} = \frac{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}{\sqrt{1^2+1^2+1^2}}$$

$k=2,$

$$P_2 = u_2 - (u_2^T q_1) q_1$$

$$= \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} - \left(\begin{bmatrix} -3 & 0 & 6 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{3}} \right) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{3}}$$

$$= \begin{pmatrix} -4 \\ -1 \\ 5 \end{pmatrix}$$

Step 2:

$$Q = [q_1 \ q_2] = \begin{pmatrix} 1/\sqrt{3} & -4/\sqrt{42} \\ 1/\sqrt{3} & -1/\sqrt{42} \\ 1/\sqrt{3} & 5/\sqrt{42} \end{pmatrix}$$

Step 3:

$$R = \begin{pmatrix} u_1^T q_1 & u_2^T q_1 \\ u_1^T q_2 & u_2^T q_2 \end{pmatrix}$$

$$u_1^T q_1 = (1 \ 1) \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} = \sqrt{3}$$

$$u_2^T q_1 = \sqrt{3}$$

$$= \begin{pmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{pmatrix}$$

$$u_2^T q_2 = \sqrt{42}$$

Step 4:

$$R x = Q^T b$$

$$\Rightarrow \begin{pmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 2/\sqrt{3} \\ 10/\sqrt{42} \end{pmatrix}$$

$$\textcircled{1} \rightarrow \sqrt{42} a_1 = 10\sqrt{42} \quad \left. \begin{array}{l} \textcircled{1} \rightarrow \sqrt{42} a_1 = 10\sqrt{42} \\ \textcircled{2} \rightarrow \sqrt{3} a_0 + \sqrt{3} a_1 = 2\sqrt{3} \end{array} \right\} \text{solve} \rightarrow a_0 = \frac{3}{7}, a_1 = \frac{5}{21}$$

$$\textcircled{2} \rightarrow \sqrt{3} a_0 + \sqrt{3} a_1 = 2\sqrt{3}$$

$$p_1(x) = a_0 + a_1 x$$

$$= \frac{3}{7} + \frac{5}{21} x$$

6.2

Example:

$$a_0 + a_1 + a_2 = 2$$

$$a_0 + 2a_1 + 4a_2 = 3$$

$$a_0 + 3a_1 + 9a_2 = 6$$

$$a_0 + 4a_1 + 16a_2 = 4$$

Find approximate value of a_0, a_1, a_2 using QR decomposition.

$\Rightarrow$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{pmatrix}$$

$$b = \begin{pmatrix} 2 \\ 3 \\ 6 \\ 4 \end{pmatrix}$$

$$x = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}$$

Step 1:

$k=1$,

$$P_1 = u_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$q = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{4}} = \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix}$$

$$P_k = U_k - \sum_{i=1}^{k-1} (u_k^T q_i) q_i$$

$$q_k = \frac{P_k}{|P_k|}$$

$k=2$

$$P_2 = u_2 - (u_2^T q_1) q_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} - \left[(1 \ 2 \ 3 \ 4) \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} \right] \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} - 5 \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} = \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix}$$

$$q_2 = \frac{P_2}{|P_2|} = \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix} \frac{1}{\sqrt{5}}$$

$k=3$

$$P_3 = u_3 - (u_3^T q_1) q_1 - (u_3^T q_2) q_2$$

$$= \begin{pmatrix} 1 \\ 4 \\ 9 \\ 16 \end{pmatrix} - \left((1 \ 4 \ 9 \ 16) \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} \right) \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} - \left((1 \ 4 \ 9 \ 16) \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix} \frac{1}{\sqrt{5}} \right) \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix} \frac{1}{\sqrt{5}}$$

$$= \begin{pmatrix} 1 \\ 4 \\ 9 \\ 16 \end{pmatrix} - \begin{pmatrix} 7.5 \\ 7.5 \\ 7.5 \\ 7.5 \end{pmatrix} - \begin{pmatrix} -87.5 \\ -42.5 \\ 42.5 \\ 87.5 \end{pmatrix} = \begin{pmatrix} 31 \\ 9 \\ -11 \\ -29 \end{pmatrix}$$

$$q_3 = \frac{P_3}{|P_3|} = \begin{pmatrix} 0.5 \\ -0.5 \\ -0.5 \\ 0.5 \end{pmatrix}$$

$$Q = (q_1 \ q_2 \ q_3) = \begin{pmatrix} 0.5 & -1.5/\sqrt{5} & 0.5 \\ 0.5 & -0.5/\sqrt{5} & -0.5 \\ 0.5 & 0.5/\sqrt{5} & -0.5 \\ 0.5 & 1.5/\sqrt{5} & 0.5 \end{pmatrix}$$

$$R = \begin{pmatrix} u_1^T q_1 & u_2^T q_1 & u_3^T q_1 \\ 0 & u_2^T q_2 & u_3^T q_2 \\ 0 & 0 & u_3^T q_3 \end{pmatrix} = \begin{pmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{pmatrix}$$

$$Rx = Q^T b$$

$$\Rightarrow \begin{pmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ -1.5/\sqrt{5} & -0.5/\sqrt{5} & 0.5 & 1.5/\sqrt{5} \\ 0.5 & -0.5 & -0.5 & 0.5 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 7.5 \\ \frac{9}{2\sqrt{5}} \\ -1.5 \end{pmatrix}$$

$$\textcircled{a} \ 2a_2 = -1.5 \Rightarrow a_2 = -0.75$$

$$\textcircled{b} \ \sqrt{5}a_1 + 5\sqrt{5}a_2 = \frac{9}{2\sqrt{5}} \Rightarrow a_1 = 4.65$$

$$\textcircled{c} \ 2a_0 + 5a_1 + 15a_2 = 7.5 \Rightarrow a_0 = -2.25$$

$$\begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} -2.25 \\ 4.65 \\ -0.75 \end{pmatrix}$$